

⇒ Writing output of a digital system using minterms:-

(a) Truth Table for a digital system is as follows

	A	B	output = $y = f(A, B)$
$m_0 \quad \bar{A}\bar{B} \leftarrow$	0	0	1
$m_1 \quad \bar{A}B \leftarrow$	0	1	0
$m_2 \quad A\bar{B} \leftarrow$	1	0	1
$m_3 \quad AB \leftarrow$	1	1	0

Find $y = f(A, B) = ?$

$$\Rightarrow y = (\bar{A}\bar{B} \cdot 1) + (\bar{A}B \cdot 0) + (A\bar{B} \cdot 1) + (AB \cdot 0)$$

$$= \bar{A}\bar{B} + A\bar{B} = \bar{B} (A + \bar{A}) = \bar{B}$$

$(m_0 + m_2) = \Sigma m(0, 2)$

(b)

D	A	B	C	$y = f(A, B, C)$
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	1
5	1	0	1	0
6	1	1	0	0
7	1	1	1	1

$y = f(A, B, C) ?$

$y = f(A, B, C)$

$$= (\bar{A}\bar{B}C \cdot 1) + (\bar{A}BC \cdot 1) + (A\bar{B}C \cdot 1) + (ABC \cdot 1)$$

$$= \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$$
~~$$= \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$~~

$$= \Sigma m(0, 3, 4, 7)$$

⇒ concept of don't care (X):-

(a) $y = f(A, B, C) ?$

	A	B	C	y
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	X	1
5	1	0	X	0
6	1	1	0	1
7	1	1	1	0

MSB (A) LSB (C)

$X=0 \Rightarrow 100 (A\bar{B}\bar{C})$

$X=1 \Rightarrow 101 (A\bar{B}C)$

$$\Rightarrow Y = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$$

$$= \Sigma m(0, 3, 4, 5, 6)$$

(*) Sum of all minterms

(Q) $y = f(A, B, C, D)$ (MSB) (LSB) = ?

D	A	B	C	y
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	0
7	1	1	1	1

$$y = (\overline{A}\overline{B}\overline{C} \cdot 1) + (\overline{A}\overline{B}C \cdot 1) + \overline{A}BC + (\overline{A}\overline{B}\overline{C} \cdot 0)$$

$$+ (\overline{A}\overline{B}C \cdot 0)$$

$$\Rightarrow \text{The given fn. has 4 Variables. So, for 3 Vars: You sum them.}$$

(Q) $y = f(A, B, C, D) = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D}$

Find min-terms?

$$\Rightarrow m_0 + m_1 + m_6 + m_7 + m_9 + m_{10} + m_{14} + m_{15}$$

$$\Sigma m(0, 1, 6, 7, 9, 10, 14, 15)$$

Maxterm:- A logical OR operation including all variables, where each variable can be itself (or) its complement.

→ o/p of Maxterm function is 0.

→ Represented by M.

→ 0 → Normal form, 1 → Complemented form.

2-Variable:-

D	A	B	Max Term	Exp
0	0	0	$\overline{A} + \overline{B}$	M_0
1	0	1	$\overline{A} + B$	M_1
2	1	0	$A + \overline{B}$	M_2
3	1	1	$A + B$	M_3

Eg:- $f(w, x, y, z) = \overline{w} + x + y + \overline{z}$

Expression for Maxterm?

$$\Rightarrow 1001 = 1x2^3 + 0x2^2 + 0x2^1 + 1x2^0$$

$$= 1 + 8 = 9 = M_9$$

(*) n-Variables = 2^n Maxterms.

→ Writing o/p of a digital system using maxterms:-

(Q) Truth table for a D.S is as follows

A	B	$y = f(A, B)$
0	0	1
0	1	0
1	0	1
1	1	0

$\rightarrow A+B$
 $\rightarrow A+\bar{B}$
 $\rightarrow \bar{A}+B$
 $\rightarrow \bar{A}+\bar{B}$

Find $y = f(A, B)$

A: $y = \underline{AB} + \underline{\bar{A}\bar{B}} = \underline{B}$. (minterms)

$$y = (A+B+1) \cdot (A+B+0) \cdot (\bar{A}+B+1) \cdot (\bar{A}+B+0)$$

$$= 1 \cdot (A+B) \cdot 1 \cdot (\bar{A}+\bar{B})$$

$$= (A+B) \cdot (\bar{A}+\bar{B}) = \underline{AB} + \underline{\bar{A}\bar{B}} + \underline{B}$$

$M_1 + M_3 = \bar{B}$
 $M_1 + M_3 = (1, 0, 1, 0) = \bar{B}$

$\Rightarrow y = \pi(M_1, M_3)$

(Q) Find $y = f(A, B, C)$?

D	A	B	C	$y = f(A, B, C)$
0	0	0	0	1
1	0	0	1	0 ✓
2	0	1	0	0 ✓
3	0	1	1	1
4	1	0	0	1
5	1	0	1	0 ✓
6	1	1	0	0 ✓
7	1	1	1	1

A: $y = \pi M(2, 4, 5, 3) = \pi M(2, 3, 4, 5)$

$$y = (\bar{C}+A+B) \cdot (C+\bar{B}+A) \cdot (C+B+\bar{A}) \cdot (C+\bar{B}+\bar{A})$$

Concept of Don't Care (X):

(Q) Find $y = f(A, B, C)$ if Truth Table is given

	A	B	C	$y = f(A, B, C)$
0	0	0	0	1
1	0	0	1	0 ✓
2	0	1	0	0 ✓
3	0	1	1	0
4	1	0	0	1 ✓
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

A: $y = (A+B+\bar{C}) \cdot (A+B+C) \cdot (\bar{A}+B+\bar{C}) \cdot (\bar{A}+B+C)$

$\Rightarrow (X+\bar{C}) \cdot (X+C)$

$\Rightarrow X + C\bar{C} = \underline{X}$

$= (A+B+\bar{C}) \cdot (A+B+C) \cdot (\bar{A}+B) \cdot (\bar{A}+B)$

(★) Product of all Max terms :-

(Q) Find $y = f(A, B, C, D)$ if Truth Table is given below?

0	A	B	C	$y = f(A, B, C)$
0	0	0	0	1
1	0	0	1	0 ✓
2	0	1	0	0 ✓
3	0	1	1	1
4	1	0	0	0 ✓
5	1	0	1	0 ✓
6	1	1	0	0 ✓
7	1	1	1	1

A. $y = (A+B+\bar{C}) \cdot (A+\bar{B}+C) \cdot (\bar{A}+B+C) \cdot (\bar{A}+B+C+D)$ (M₈)

$(\bar{A}+B+\bar{C}+\bar{D})$ (M₁₄)

$= (M_2 \cdot M_3) \cdot (M_4 \cdot M_5) \cdot M_8 \cdot M_{11} \cdot (M_{12} \cdot M_{13})$

$= \pi M (2, 3, 4, 5, 8, 11, 12, 13)$

(Q) $y[A, B, C, D] = (A+B+C) (\bar{A}+B+C) (\bar{A}+B+C+\bar{D})$

(Pos)

A. $y = (M_0 \cdot M_1) \cdot (M_8 \cdot M_9) \cdot (M_9)$

$= \pi M (0, 1, 8, 9)$

(Q) $f(A, B, C) = \sum m(0, 2, 4)$: Max-term?

A. 3 Variables $\Rightarrow$ 7 decimals.

$\pi M (1, 3, 5, 6, 7)$

(Q) $F(A, B, C, D) = \pi M (6, 2, 3, 5, 7)$: Min-term?

A. 4 Variables $\Rightarrow$ 15 Variables

$f(A, B, C, D) = \sum m (1, 4, 6, 8, 9, 10, 11, 12, 13, 15)$

$\Rightarrow$ Sum of Product and Product of Sum :-

(i) SOP :- Product terms "OR"ed together.

Eg:- $ABC + \bar{A}\bar{B}, \bar{A}\bar{B} + BCD$

(ii) POS :- Sum terms "AND"ed together.

Eg:- $(A+B+C)(D+E), A+B+C$

$\Rightarrow$ Standard / Canonical SOP/POS form :-

All the Variables should be Present in each Product (or) Sum term. [either in normal form (or) Complemented].

Ex:- $f(A, B, C)$ is given as $m = (0, 2, 4)$

- $(A+B)(B+\bar{C})$
- $(A+B+C)(A+B+\bar{C})(A+B+\bar{C})$
- $AB+B\bar{C}$
- $AB\bar{C} + AB\bar{C} + \bar{A}BC$

In which of them is standard SOP/POS?

- Not POS
- Standard POS
- Not SOP
- Standard SOP

(a) For given function, write min. terms and max terms. Then write standard SOP and POS form.

(a) $f(A, B, C) = A + \bar{B}C$

(b) $f(A, B, C) = AB + \bar{A}C$

(c) $f(A, B, C) = A + B + \bar{C}$

(a) $f(A, B, C) = A + \bar{B}C$

Min terms $\rightarrow 100, 101, 110, 111, 001, 101$

$\bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} + ABC$

$= A[\bar{B}\bar{C} + \bar{B}C + B\bar{C} + BC] = A$

Standard SOP: $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$

$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$ (SOP)

$= 100, 101, 110, 111, 001, 010, 011, 100$

(Min terms) $= 1, 4, 5, 6, 7, 1 = \sum m(1, 4, 5, 6, 7)$

((Max terms) $= \prod M(0, 2, 3)$

$(010, 001, 010, 000, 000, 001, 010, 011)$

$\Rightarrow f(A, B, C) = (\bar{A} + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + C)$ (Standard POS)

Standard POS Method - 2:-

$f(A, B, C) = A + \bar{B}C$

$= (A + \bar{B})(A + C)$

$+ \bar{A}B + \bar{A}C + \bar{A}B + \bar{A}C = (\bar{A}B + \bar{A}C) + (A + \bar{B})(A + C)$

$= (\bar{A} + \bar{B} + C)(\bar{A} + B + \bar{C})(A + B + C)(A + \bar{B} + C)$

$\bar{A}B + \bar{A}C = (\bar{A} + \bar{B} + C)(\bar{A} + B + \bar{C})(A + B + C)(A + \bar{B} + C)$

$+ \bar{A}B + \bar{A}C + \bar{A}B + \bar{A}C = (\bar{A}B + \bar{A}C) + (A + \bar{B})(A + C)$

$\bar{A}B + \bar{A}C = (\bar{A} + \bar{B} + C)(\bar{A} + B + \bar{C})(A + B + C)(A + \bar{B} + C)$

$(110, 101, 010, 001, 011, 000, 111)$

$$(b) f(A, B, C) = AB + \bar{A}C$$

$$\Rightarrow \text{Standard SOP:- } f(A, B, C) = AB + \bar{A}C$$

$$= AB\bar{C} + ABC + \bar{A}BC + \bar{A}\bar{B}C$$

$$110, 111, 011, 001$$

$$\rightarrow \text{Min terms} = \sum m(1, 3, 6, 7)$$

$$\text{Max terms} = \prod M(0, 2, 4, 5) \Rightarrow 000, 010, 100, 101$$

$$\Rightarrow \text{Standard Pos:- } F(A, B, C) = (A+B+C) \cdot (A+\bar{B}+C) \cdot (\bar{A}+B+C) \cdot (\bar{A}+B+\bar{C})$$

$$(c) f(A, B, C) = (A+B+\bar{C}) \rightarrow (\text{Standard Pos})$$

$$\Rightarrow \text{Standard SOP:-}$$

$$f(A, B, C) = \underline{ABC} + \underline{A\bar{B}C} + \underline{AB\bar{C}} + \underline{A\bar{B}\bar{C}} + \underline{\bar{A}BC} + \underline{\bar{A}\bar{B}C} + \underline{AB\bar{C}} + \underline{\bar{A}B\bar{C}} + \underline{A\bar{B}\bar{C}} + \underline{\bar{A}\bar{B}\bar{C}} + \underline{ABC} + \underline{\bar{A}BC} + \underline{A\bar{B}C} + \underline{\bar{A}\bar{B}C}$$

$$= \underline{ABC} + \underline{A\bar{B}C} + \underline{AB\bar{C}} + \underline{\bar{A}\bar{B}\bar{C}} + \underline{A\bar{B}C} + \underline{\bar{A}\bar{B}\bar{C}} + \underline{\bar{A}BC} + \underline{\bar{A}\bar{B}C}$$

$$\text{min terms} \Rightarrow \sum m(111, 000, 110, 100, 010, 101, 011)$$

$$\Rightarrow \sum m(0, 2, 3, 4, 5, 6, 7)$$

$$\Rightarrow \text{Max terms} \Rightarrow \prod M(1)$$

END OF
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